ECON 0100 | HW C1 | Externalities

Solutions

Answers and work are shown in the red blocks.

Q1 | Malodorous Byproducts

Butterbeer is beloved by Hogwarts students and teachers alike and is widely available at wizarding shops. But the beverage has the negative effect of generating displeasing odors from its imbibers. Model the private market for butterbeer using private and social cost and benefit curves. Indicate on your graph the private equilibrium quantity and social equilibrium quantity.

Preferences for butterbeer and its marginal cost can be represented by the following relationships.

$$D : P_B = 100 - \frac{1}{2}Q_D \qquad S : P_S = 20 + \frac{1}{2}Q_S$$

Prices are in galleons and quantity is in bottles. The odor imposes a cost of 20 galleons per bottle on everyone nearby.

Marginal Social Cost: _____

Market Equilibrium Price ($\hat{P}$): _____

Market Equilibrium Quantity ($\hat{Q}$): _____

Socially Efficient Quantity (Q^*): _____

Deadweight Loss (DWL): _____

Marginal Social Cost: $P = 40 + \frac{1}{2}Q$. Market Equilibrium Price: 60 galleons. Market Equilibrium Quantity: 80 bottles. Socially Efficient Quantity: 60 bottles. Deadweight Loss: 200 galleons.

Demand is the marginal private benefit (MPB) and supply is the marginal private cost (MPC). The odor adds 20 of cost to every bottle, borne by people nearby, so the marginal social cost sits 20 above supply:

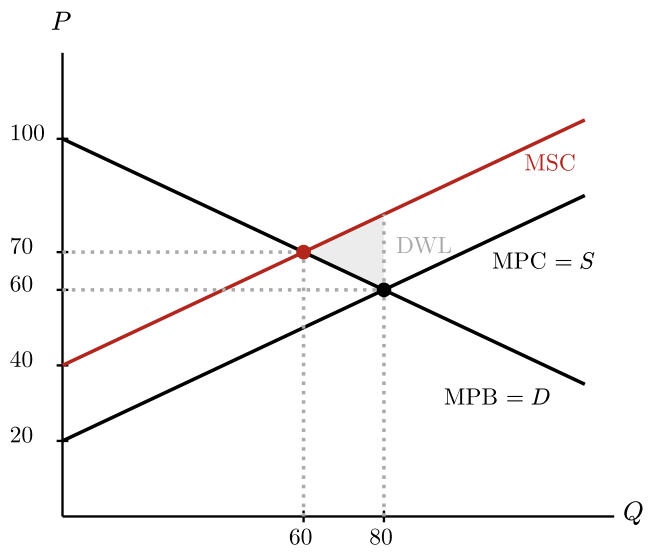
$$\text{MSC} = (20 + \tfrac{1}{2}Q) + 20 = 40 + \tfrac{1}{2}Q$$

The market ignores the odor, so it clears where demand meets supply:

$$100 - \tfrac{1}{2}Q = 20 + \tfrac{1}{2}Q \quad \Rightarrow \quad Q = 80, \quad P = 60$$

The efficient quantity counts the odor, so demand meets MSC:

$$100 - \tfrac{1}{2}Q = 40 + \tfrac{1}{2}Q \quad \Rightarrow \quad Q = 60, \quad P = 70$$



Bottles 61 through 80 cost society more (MSC) than buyers value them (MPB), yet the market makes them. The deadweight loss is the triangle between MSC and demand over those 20 bottles. Its height at $Q = 80$ is $\text{MSC} - \text{MPB} = 80 - 60 = 20$:

$$\text{DWL} = \tfrac{1}{2} \times 20 \times (80 - 60) = 200$$

Q2 | Covid Vaccine

Lets model Covid and assume the following.

1. There is a linear marginal private benefit curve for the vaccine.
2. The vaccine carries with it a positive externality.
3. There is a linear upward sloping supply curve for the vaccine. The supply curve for the vaccine is a little funny, and we'll talk about imperfect close substitutes in the next few weeks. But we'll just assume it's there.

Use a graph to discuss the welfare in the market and any potential market failure.

The graph below shows the market. The upper downward-sloping curve is marginal social benefit and the lower one is marginal private benefit. Supply is both the private and the social cost. The lettered regions are for the questions.

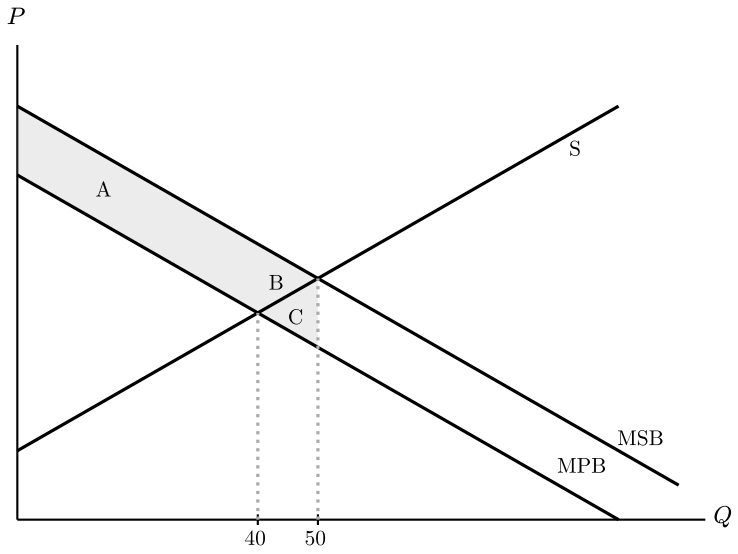


Figure 1: Covid vaccine market with MPB, MSB and S, and regions A, B and C

Market Quantity ($\hat{Q}$): _____ Socially Efficient Quantity (Q^*): _____

Deadweight Loss region (letter): _____

Market Quantity: 40 vaccinations. Socially Efficient Quantity: 50 vaccinations. Deadweight Loss region: B.

Buyers weigh only their own benefit, so the market clears where MPB meets supply. On the graph MPB is $P = 100 - Q$ and supply is $P = 20 + Q$:

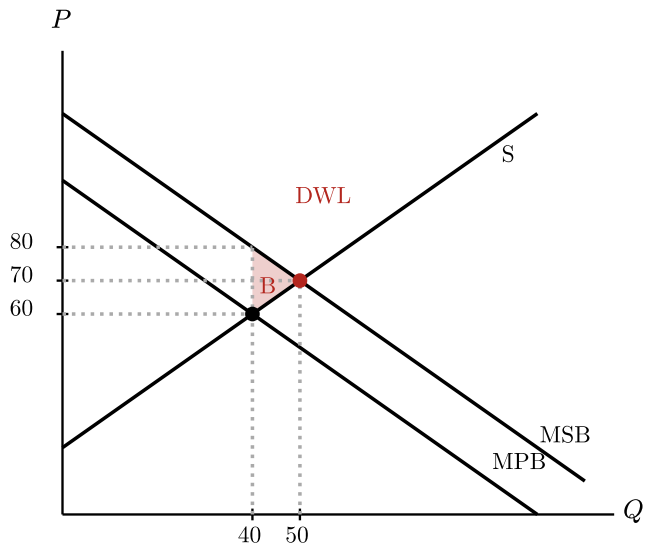
$$100 - Q = 20 + Q \quad \Rightarrow \quad Q = 40, \quad P = 60$$

The efficient quantity counts the benefit to everyone else, so MSB, $P = 120 - Q$, meets supply:

$$120 - Q = 20 + Q \quad \Rightarrow \quad Q = 50, \quad P = 70$$

The vaccinations from 40 to 50 are worth more to society (MSB) than they cost (S), but they never happen. The lost surplus is the triangle between MSB and supply over those units: region B, worth $\frac{1}{2} \times 20 \times 10 = 100$.

Region A is the external benefit the first 40 vaccinations already give to others; it is gained, not lost. Region C lies below supply, under cost, so it is not surplus at all.



Why does the market provide too few vaccinations?

a) Each vaccination costs society more to produce than it costs the seller.

b) Each vaccination is worth more to society than to the person buying it, so at the market quantity marginal social benefit is still above marginal cost.

c) Buyers value each vaccination less than sellers' marginal cost, so too few buyers enter the market.

d) The market quantity is already socially efficient, so no vaccinations are missing.

b)

The person getting vaccinated also protects the people around them, but does not count that benefit when deciding. At the market quantity of 40, MSB is 80 while marginal cost is only 60, so one more vaccination would add 20 of surplus. a) describes an external cost, the butterbeer story, not this one. c) is wrong because at $Q = 40$ the buyer's value equals marginal cost; buyers are not underpaying, they are leaving out others' benefit. d) is wrong because the market stops 10 short of the efficient 50.